Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)	Vol = $\frac{4}{3}\pi \times (0.938)^3$ (1) substitution either in mm or cm or m Vol = <u>3.457 or</u> <u>3.46 or 3.5</u> [cm^3] (1) % unc in $V = 3 \times 0.02 \times \frac{100}{18.76}$ = 0.32 [%] seen **or** % unc in r = 0.1 % so if you cube r then 3 × uncertainty so 0.32 [%] (1)	1	1 1		3	3	3
		(ii)	% unc in mass = $0.5 \times \frac{100}{26.3}$ **or** shown = 1.9 [%] (1) Density = $\frac{26.3}{3.46\ \textbf{ecf}}$ = 7.61 [$g\,cm^{-3}$] (1) Total % unc = 2.2 [%] **ecf** on % unc in mass (1) Density shown as 7.6 ± 0.2 [$g\,cm^{-3}$] **ecf** Accept 7.61 ± 0.17 (1) No more than 2 sig figs for the uncertainty.		4		4	4	4
	(b)		**For the 1st mark:** – % unc in vol = 2.9 [%] or greater **and** [%] unc in mass same as for Tomos / unchanged – % uncertainty in density = 4.8 [%] calculated – [absolute] uncertainty for Tomos = 0.01 [cm^3] **and** [%] unc in mass is the same – calculate [absolute] uncertainty for Jerry = 0.37 or 0.4 [$g\,cm^{-3}$] **2nd mark:** So [absolute] unc in density great<u>er</u> for Jerry **or** [absolute] unc in density low<u>er</u> for Tomos			2	2	1	2